



# OPEN FDTD-based design of high quality factor quantum dot photonic crystal nanolaser for next-generation nanotechnologies

Ali Farmani<sup>✉</sup> & Anis Omidniaee

Photonic crystal cavities engineered for high-performance terahertz (THz) emission offer a promising route toward compact quantum-photonic platforms. Here, we introduce a hybrid indium phosphide/zinc oxide (InP/ZnO) quantum-dot photonic crystal laser that exploits the wide bandgap of ZnO ( $\approx 3.37$  eV) and its highly scalable nanorod and thin-film growth modes to sustain strongly confined optical states. By systematically analyzing the quality factor under variations in temperature-dependent dispersion and radius-to-lattice constants, we demonstrate a substantial enhancement in spontaneous emission and optical pumping efficiency. Our optimized structure achieves quality factors up to 1600 for InP and further improved performance in the InP/Al<sub>2</sub>O<sub>3</sub>/ZnO hybrid gain medium. Finally, we route the photonic crystal laser output into quantum logic-gate architectures and quantify its angle-dependent rotation dynamics and probability distributions, highlighting its feasibility for next-generation quantum laser processing.

**Keywords** Photonic crystal laser, ZnO quantum dots, Terahertz photonics, High-Q cavities, Quantum logic gates

Photonic crystal nanostructures have emerged as a foundational platform for nanoscale light generation, offering unprecedented control over optical confinement, modal engineering, and material-photon coupling strength. Early demonstrations of membrane-based photonic devices showed that strong vertical and lateral confinement could be achieved even on non-native substrates, enabling large-area standing-wave modes with exceptionally narrow far-field divergence<sup>1–4</sup>. These developments led to nanolasers with ultralow thresholds—down to a few kilowatts per square centimeter—using colloidal quantum dots embedded in high-index photonic lattices, where smooth dielectric surfaces and precisely filled air holes were essential for boosting quality factors and suppressing scattering losses<sup>5–7</sup>.

Progress in cavity engineering accelerated with the introduction of twisted photonic-crystal nanocavities and double-layer architectures, where small rotation angles between stacked lattices produced tunable, high-Q resonances within sub-wavelength mode volumes<sup>8–11</sup>. Such designs enabled room-temperature single-mode emission with thresholds near 1 kW/cm<sup>2</sup> and continuous-wave operation at telecom wavelengths with threshold currents approaching 10  $\mu$ A. Despite these advances, traditional III–V systems—particularly InP- and GaAs-based quantum dots—suffered from carrier leakage, limited injection efficiency, and spectral broadening arising from lattice mismatch during Stranski–Krastanov growth. Even modest mismatch values introduced significant inhomogeneity in dot size distribution, necessitating complex optimization of epitaxial layer thickness and indium incorporation techniques<sup>12–14</sup>. While lateral pumping and proton implantation reduced threshold currents to tens of milliamperes, thermal degradation and limited spatial overlap between the cavity mode and the quantum emitter continued to pose fundamental challenges.

Parallel efforts in colloidal lead chalcogenide quantum dots explored surface-emitting nanolasers with enhanced nonlinear optical behavior. However, their performance was hindered by rapid Auger recombination and trap-induced nonradiative losses. Hybridization with ZnO nanocrystals successfully suppressed trap states, extending the Auger lifetime nearly fivefold and halving the spontaneous-emission threshold at 1650 nm. Core–shell ZnSe/CdSe quantum dots further improved performance in circular Bragg resonators, achieving confinement factors approaching unity and Purcell enhancements exceeding 20, along with threshold reductions surpassing 70% compared with vertical-cavity systems. Nevertheless, the inherent randomness of self-assembled dots limited spectral uniformity and spatial positioning, restricting deterministic integration with nanocavities.

Department of Electronics Engineering, Lorestan University, Khorram Abad, Iran. ✉email: farmani.a@lu.ac.ir

Although buried photonic defects and Moiré photonic graphene layers offered routes toward controlled localization and flat-band formation, practical integration remained challenging due to insufficient field overlap and sensitivity to structural perturbations<sup>15–17</sup>.

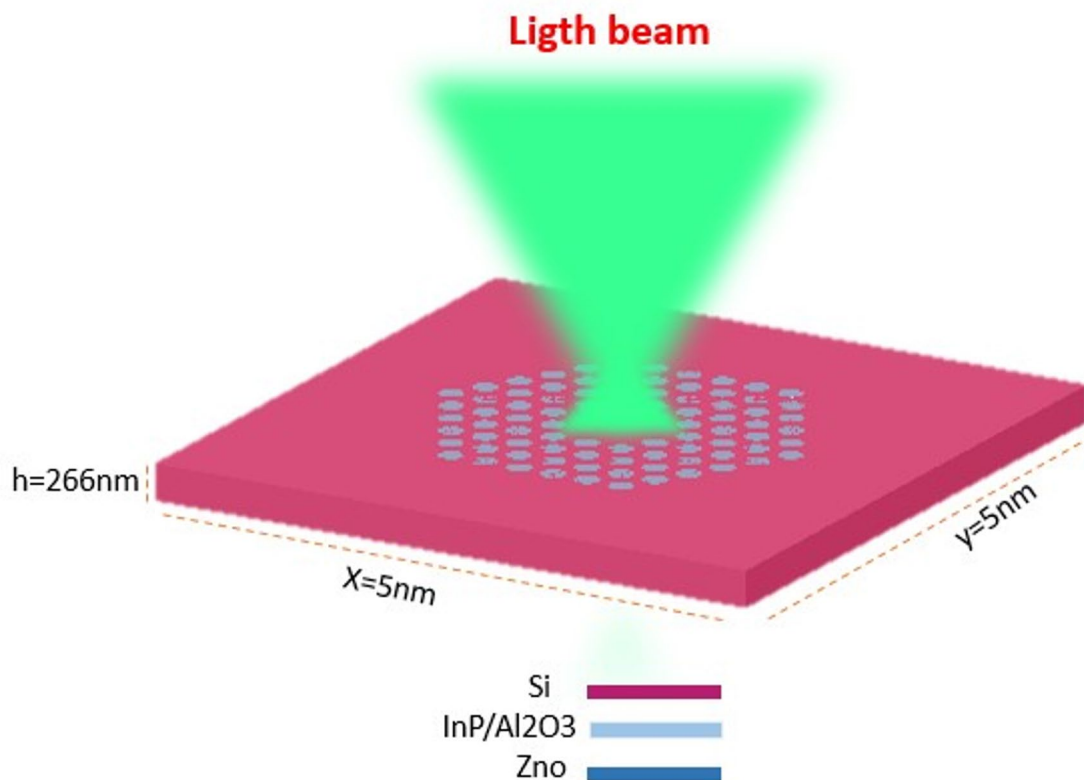
These cumulative advances point toward a clear trend: hybrid, wide-bandgap semiconductor platforms—particularly those incorporating ZnO nanorods, nanowires, or thin films—provide a path to overcoming the long-standing limitations of traditional III–V quantum-dot nanolasers. ZnO's wide bandgap ( $\sim 3.37$  eV), strong excitonic stability, and adaptability to various nanostructured morphologies make it uniquely suited for sustaining tightly confined photonic modes under strong excitation<sup>12,18–20</sup>.

When combined with indium phosphide and dielectric intermediaries such as  $\text{Al}_2\text{O}_3$ , hybrid gain media can suppress nonradiative pathways, stabilize emission, and significantly enhance the spontaneous-emission rate.

Building upon this rich research landscape, the present work introduces a high-quality-factor InP/ZnO quantum-dot photonic crystal laser cavity, optimized through systematic tuning of temperature-dependent dispersion and radius-to-lattice-constant ratios. Our design achieves Q-factors approaching 1600 in hybrid configurations and demonstrates strong terahertz-range confinement suitable for robust optical pumping. Crucially, we extend the functionality of this photonic crystal laser by coupling its output into quantum logic-gate architectures, enabling precise control over rotation angles and probability distributions—highlighting its potential as a compact, scalable light source for emerging quantum photonic computation.

### Desing and structure

A photonic crystal-based nanolaser was developed utilizing the finite difference time domain numerical (FDTD) solution method. The FDTD simulations in this study were performed using Tidy3D Python FDTD Solutions, a commercial finite-difference time-domain software widely used for nanophotonic device modeling. The simulations employed a nonuniform mesh with a minimum resolution of 2 nm in the active photonic-crystal region to accurately capture subwavelength features. Material dispersion and absorption were included for all semiconductor and dielectric layers, including InP, ZnO, and  $\text{Al}_2\text{O}_3$ , using experimentally measured complex refractive indices. Perfectly matched layer (PML) boundary conditions were applied in all directions to prevent artificial reflections, and time-domain monitors were used to record electric-field evolution, output spectra, and quality factors. This design incorporated a hybrid gain medium consisting of a 10 nm thick layer of aluminum dioxide, which has a refractive index of 1.75, alongside a gallium arsenide oxide dielectric layer. The primary objective of this configuration was to enhance optical pumping. This structure, as seen in Fig. 1, the photonic crystal profiles in the symmetric A-state and the antisymmetric B-state is made up of a silicon substrate with a 40 nm thick, and a indum phosphide layer is formed on top of it with a 170 nm thickness, which is situated 40 nm thick zinc oxide layers and 5 nm thick  $\text{Al}_2\text{O}_3$  a layers, respectively.



**Fig. 1.** Three-dimensional perspective of the photonic crystal nanolaser architecture.

The membrane of the photonic crystal consists of a triangular lattice featuring air holes, with a lattice pitch, denoted as 'a', measuring 366 nm. In this context, a hole radius of 135.4 nm, equivalent to 0.37a, is considered. Photonic crystal nanolasers, which utilize thin semiconductor films in conjunction with single indium phosphide quantum dots, are particularly noteworthy due to their minuscule volume and ability to manipulate the magnetic field within the cavity, facilitating atom emission. This capability allows for the design of polarization-related patterns. The incorporation of silicon in these structures is advantageous due to its low cost, high integration potential, and compatibility with integrated circuit manufacturing processes. However, the optical pumping efficiency of these devices is constrained when paired with gallium arsenide and aluminum gallium arsenide materials. Research indicates that zinc oxide is a promising alternative, as it possesses a relatively high exciton binding energy of approximately 60 meV, which exceeds that of other materials<sup>12</sup>.

$$n_{ave} = \sqrt{n_b^2 (1 - FF) + n_h^2 \bullet FF} \quad (1)$$

The refractive index of the collision zone and air is represented by  $n_b, n_h$  in this relationship.

Several defect cavities have been used in photonic crystal structures. Defect structures in photonic crystals are very important for the propagation and operation of lasers, because these defects practically allow the control, confinement, and guidance of light, and the defects act like cavities, allowing light to be very strongly confined in a limited area within the photonic crystal. This confinement increases the Q (quality factor of resonance) and reduces the optical mode volume (mode volume). According to the Purcell effect The spontaneous emission rate of a light source can be significantly increased when placed in a cavity with high Q and small mode volume:

$$FP = \frac{3}{4\pi^2} \left( \frac{\lambda_c^3}{n} \right) \frac{Q}{V} \quad (2)$$

where  $(\lambda_c/n)$  is the wavelength within the material, and Q and V are the Quality Factor and mode volume (in cubic-wavelengths) of the cavity, respectively. The photonic crystal creates allowed states in the photonic band gap that do not allow light to propagate over a range of frequencies. When defects are introduced into the structure, localized optical states appear within the band gap. These states act like energy levels in doped semiconductors, and the laser frequency is tuned precisely to this defect state.

## Results and discussions

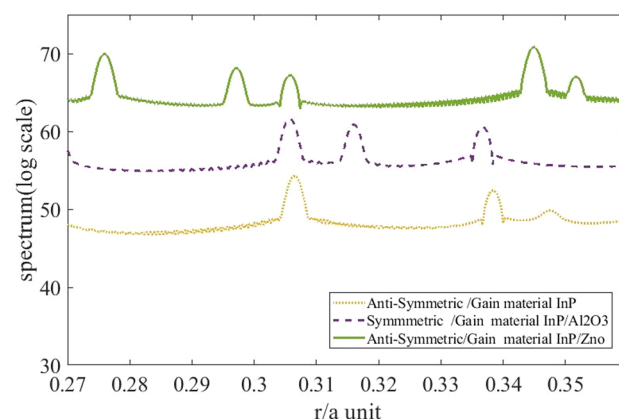
The Gaussian emission spectrum is initially found at 1210 nm and 1110 nm, respectively, in the quantum dots' recombination of the ground state and first excited electron-hole pairs. The variation in size of the self-assembled quantum dots is shown by the width of each band. The hole emission spectrum, in contrast to the wide emission bands of the quantum dot array, displays distinct lines that span the spontaneous quantum dot emission band as the ratio of the nanohole radius to the lattice constant rises. The quality factor of the hole restricts the linewidth of the modes, which increases with the quality factor ratio of 1500.

As shown in Fig. 2, the mode intensity increases as the cavity modes enter into resonance with the majority of the quantum dots, i.e., near the peak position of the quasiGaussian band.

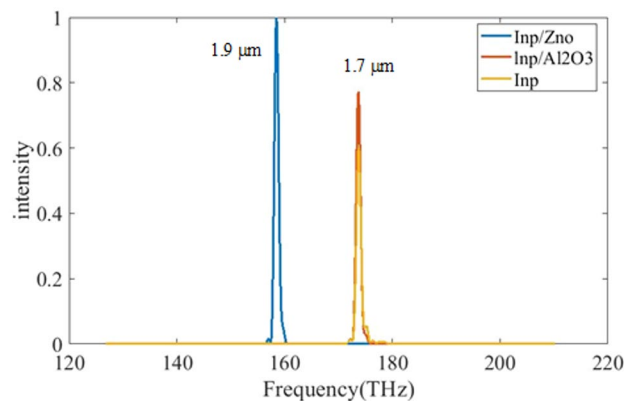
To enhance optical pumping of indium phosphide, quantum dots are positioned in the silicon interlayer. Additionally,  $\text{Al}_2\text{O}_3$  functions as a dielectric layer with a refractive index of 1.75, serving as a low refractive index material to confine the field for the gain medium; so, intensity is calculated and plotted in Fig. 3 for InP/ZnO, InP/ $\text{Al}_2\text{O}_3$  and InP.

As shown in Fig. 4. The effect of indium phosphide/zinc oxide in increasing the confinement intensity of the gain medium is also determined by their quality factor in Fig. 5, which is maximized at a frequency of 212 THz.

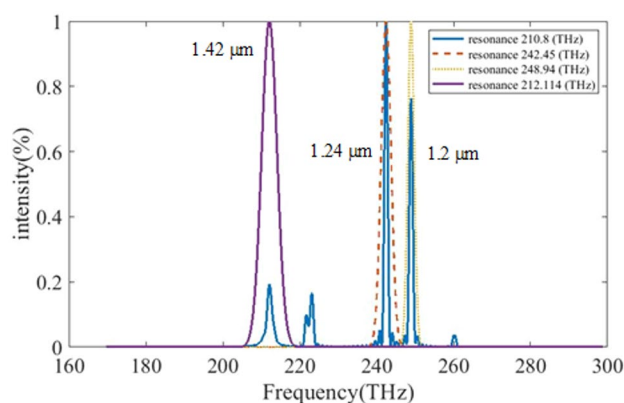
The only well-defined current injection window identified is region I, as illustrated in Fig. 5 The figure displays the curves for quantum dots with lattice constants of 366 nm, along with injection apertures of 60  $\mu\text{m}$  and 40  $\mu\text{m}$  at a temperature of 10 °C.



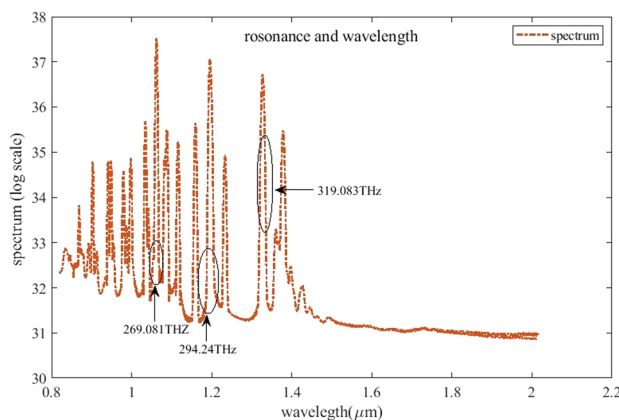
**Fig. 2.** Shows the Gaussian emission spectrum plot as a function of radius to lattice constant.



**Fig. 3.** Illustrates the emission intensity as a function of frequency, spanning from 120 to 220 terahertz, within a hybrid gain medium.

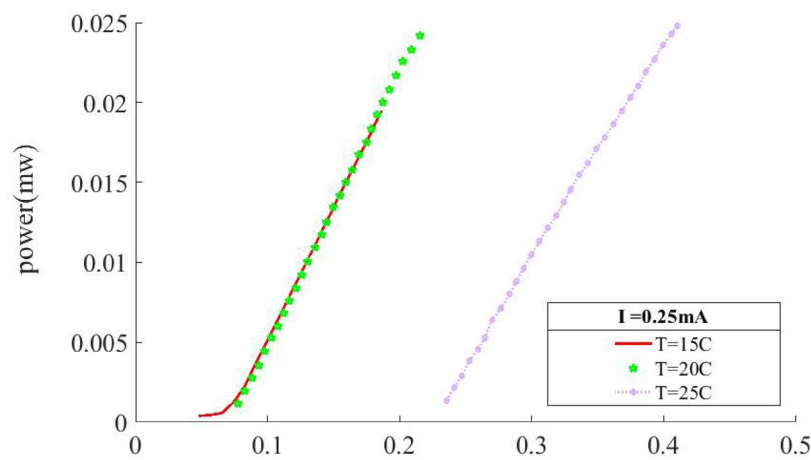


**Fig. 4.** Emission intensity vs. frequency in the range of 160 to 280 terahertz.



**Fig. 5.** The Euler spectrum, which peaks at 212 terahertz, is shown in the graph.

In Fig. 6, the power is calculated and presented. The current confinement provided by the indium phosphide and zinc oxide layers leads to a decrease in the threshold current by around 0.25 mA, even though there is an increase in the threshold current density. The device with a 0.1-micrometer injection aperture did not exhibit a reduction in threshold current. The observed increase in threshold current density is linked to the discrepancy between the gain spectra and the limitations set by the photonic crystals. Furthermore, the current density tends to rise with increasing temperature. In order to investigate the threshold current of the nanolaser, the electrical characteristics of the materials are given in Table 1.



**Fig. 6.** The output gain curve of a photonic crystal laser with current excitation in a indum phosphide / zinc oxide gain environment as a function of temperature fluctuations.

| Material                       | Type in DEVICE                | Bandgap (eV) | Relative Permittivity (εr) | Electron Mobility μn (cm <sup>2</sup> /V·s) | Hole Mobility μp (cm <sup>2</sup> /V·s) | SRH Lifetime | Radiative Coefficient B (cm <sup>3</sup> /s) |
|--------------------------------|-------------------------------|--------------|----------------------------|---------------------------------------------|-----------------------------------------|--------------|----------------------------------------------|
| ZnO                            | Semiconductor                 | 3.37         | 8.8                        | 100–200                                     | 1–10                                    | 0.1–1 ns     | —                                            |
| Al <sub>2</sub> O <sub>3</sub> | Insulator (Dielectric)        | ~8.8         | 9.0–9.5                    | —                                           | —                                       | —            | —                                            |
| InP QD                         | Semiconductor (Active region) | 1.0–1.1      | 12.4                       | 3000–5000                                   | 150–200                                 | 0.5–2 ns     | ~ 1 × 10 <sup>-10</sup>                      |
| Si                             | Semiconductor                 | 1.12         | 11.7                       | 1350                                        | 480                                     | 1–10 μs      | —                                            |

**Table 1.** Electrical characteristics of materials in the environment for electrical analysis.

The internal laser performance of the proposed InP/ZnO photonic-crystal nanolaser is strongly influenced by quantum-mechanical and carrier-dynamics phenomena. The Purcell effect enhances the spontaneous-emission rate within the high-Q cavity, where the small mode volume and strong optical confinement increase the local density of optical states. This enhancement facilitates efficient coupling of emitted photons into the cavity mode, contributing to the observed reduction in lasing threshold and improved power output. In addition, the carrier dynamics within the hybrid quantum-dot gain medium play a critical role in determining laser efficiency. Radiative recombination dominates within the confined cavity modes, producing effective stimulated emission, while non-radiative recombination is suppressed by high-quality InP/ZnO interfaces and the inclusion of thin Al<sub>2</sub>O<sub>3</sub> layers that mitigate surface and defect-assisted losses. Furthermore, Auger recombination is reduced in the wide-bandgap ZnO layers, which minimizes non-radiative lifetime shortening and enhances the internal quantum efficiency.

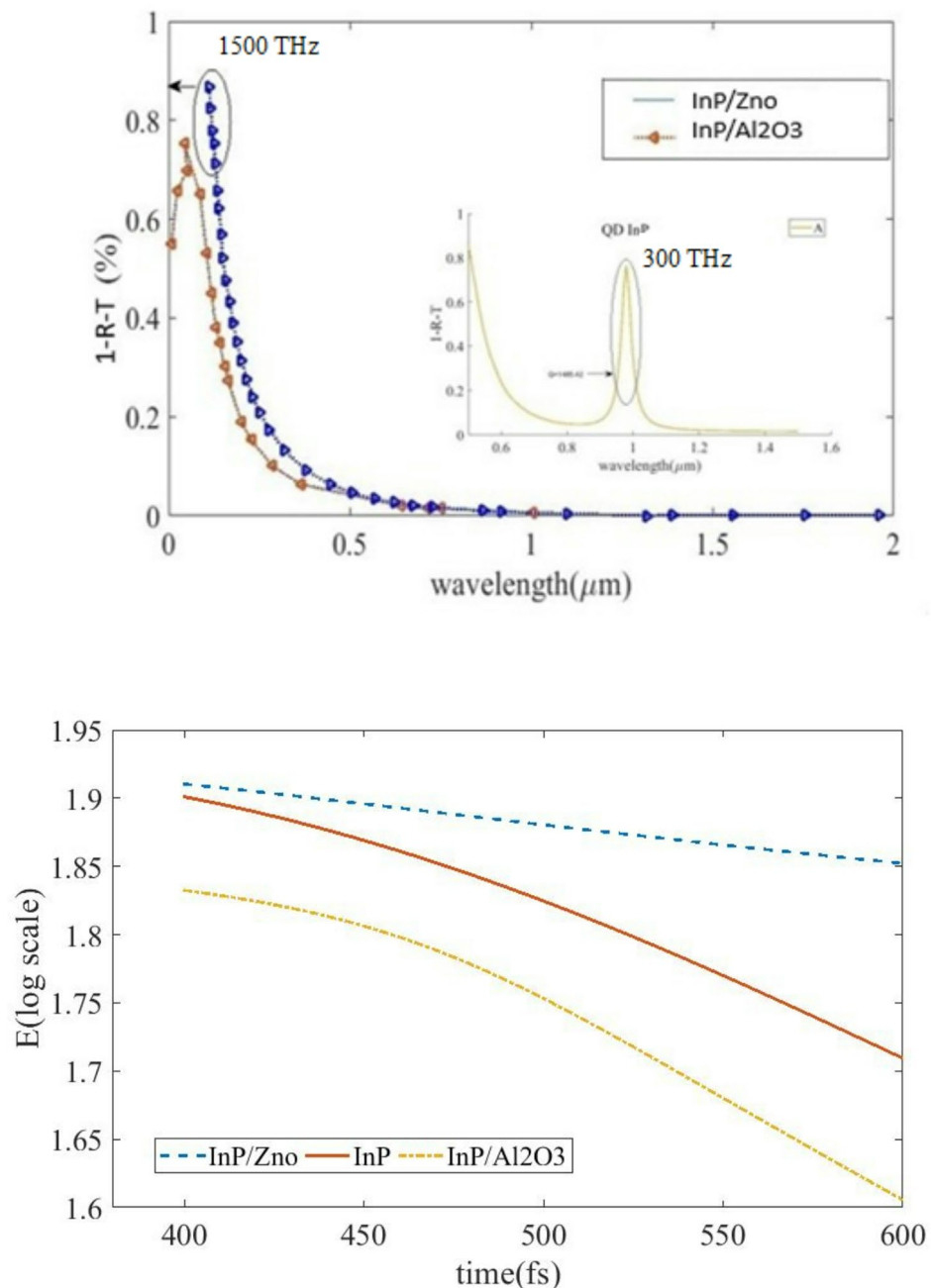
By integrating these quantum and carrier-dynamics considerations, the nanolaser achieves low-threshold operation, strong mode confinement, and stable temporal response, demonstrating the effectiveness of the hybrid photonic-crystal design for high-performance optical and quantum applications. Previous studies have demonstrated that the spontaneous emission of quantum dots can be both enhanced and suppressed depending on the cavity design and spatial positioning within the photonic crystal. Takiguchi et al. (2013) showed that buried heterostructure photonic-crystal cavities can significantly modify the local density of optical states, leading to selective enhancement of certain cavity modes while suppressing others. This effect allows for precise control over the coupling between quantum emitters and confined photonic modes, resulting in improved quality factors and reduced lasing thresholds. Our hybrid InP/ZnO H7 cavity leverages a similar principle, where the strong confinement and optimized material composition enhance radiative emission while minimizing non-radiative losses, consistent with the mechanisms reported in these foundational studies<sup>21</sup>. In addition, the strong coupling regime between a single quantum dot and a photonic-crystal nanocavity has been demonstrated by Ota et al. (2009), showing vacuum Rabi splitting in an H1 cavity. This work highlights that even single-emitter systems can experience substantial modification of emission dynamics when embedded in high-Q photonic-crystal nanocavities. While our system involves multiple quantum dots in a hybrid medium, the underlying physics remains similar: the nanocavity effectively alters the spontaneous emission rate, enhances the interaction between photons and carriers, and enables coherent emission with controlled mode properties. These observations provide a theoretical and experimental foundation for understanding the Purcell-enhanced emission and carrier dynamics in our proposed hybrid nanolaser design<sup>22</sup>.

Absorption is one of the most important factors determining the threshold, efficiency, linewidth, and output stability. Because nanolasers have very small dimensions, even a small amount of absorption can significantly change the optical and quantum behavior of the laser. Increasing absorption in the active material reduces the

Q-factor, which in turn increases the optical pumping threshold. The effect of material absorption is examined in Fig. 7a. The results show that the use of the Aluminum oxide/zinc oxide material combination shows a lower absorption of 70% compared to the indium phosphide/zinc oxide material combination. In a similar vein, the indium phosphide/quantum dot material's absorption impact has been examined in isolation throughout the wavelength range of 0.5 to 1600 nm. The quality was evaluated by measuring the absorption at 1000 nm, yielding a result of 0.8%, as demonstrated in Fig. 7a. The amount of light trapped in this area is estimated to be 1455.3 times the factor. The variations in the gain environment are depicted in Fig. 7b, indium phosphide, and aluminum dioxide coatings have all been examined. The aluminum dioxide layer has a lower peak than the variations in time with the field intensity, and it is nearly identical for all of them. the indium phosphide, arsenic, and zinc oxide layers.

In the following, Scattering diagram of hybrid active materials of photonic crystal lasers is calculated and for different cases is plotted in Fig. 8.

Laser dispersion can depend on the threshold and quality factor of the laser (as can be seen in Fig. 9).



**Fig. 7.** (a) Absorption diagram of active hybrid materials in photonic crystal lasers in wavelength of 1000 nm. (b) The curve shows the field versus time in the range of 300 to 600 femtoseconds.



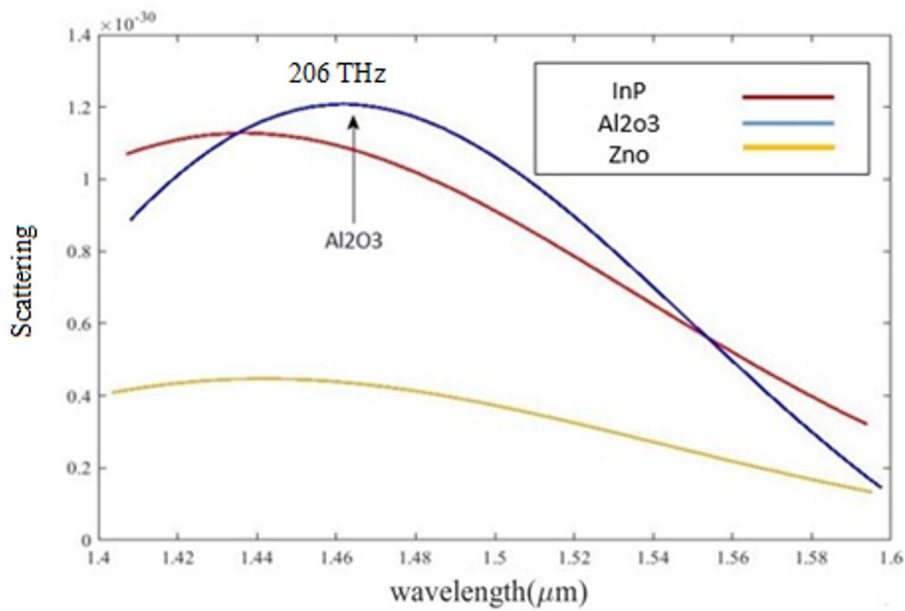


Fig. 8. Scattering diagram of hybrid active materials of photonic crystal lasers.

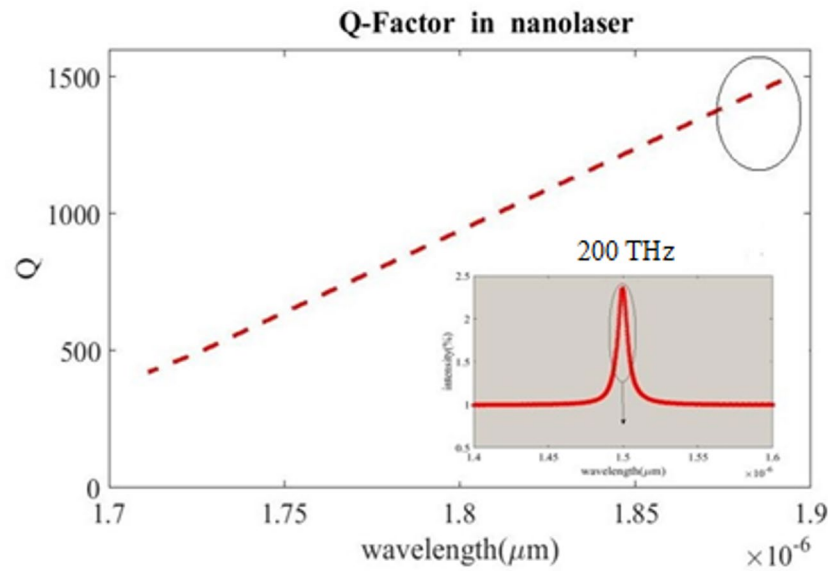
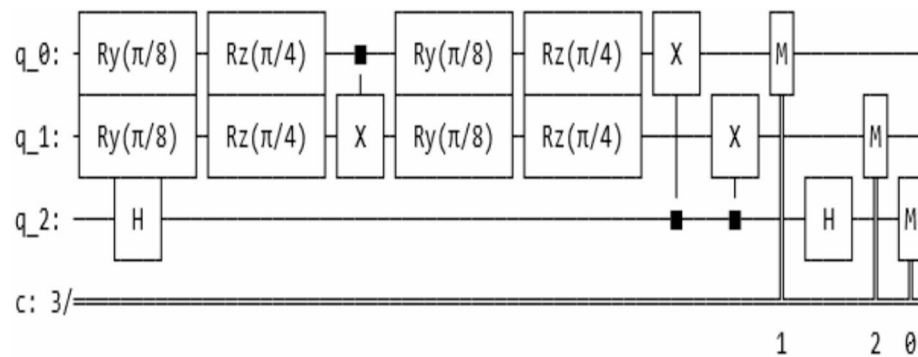


Fig. 9. Q-factor diagram of hybrid active materials of photonic crystal lasers.

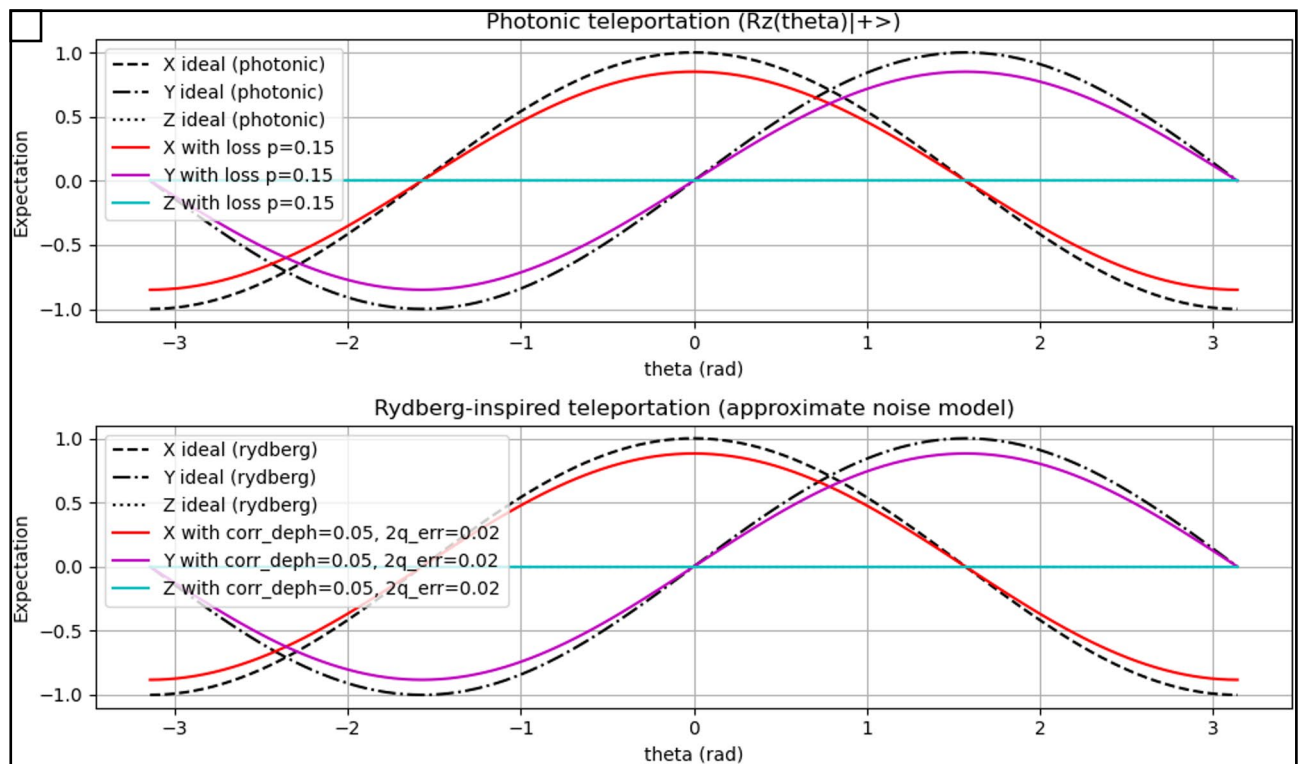
The dispersion diagram of the active hybrid materials of the photonic crystal laser shows that the combination

| Key innovation feature                                                       | Wavelength(nm) | Gain material Q |                        | Ref       |
|------------------------------------------------------------------------------|----------------|-----------------|------------------------|-----------|
| optimizing GaAs-based 1.3 μm high-performance surface-emitting lasers        | 490            | 130             | Cds                    | 23        |
| removing three holes called L3 nanocavities to achieve a high quality factor | 650            | 210             | SiN                    | 24        |
| lower threshold values for Raman silicon nanocavity lasers                   | 1530           | 170             | Si                     | 25        |
| AlN with high thermal conductivity                                           | 1420           | 85.5            | LSN PiG-TiO2-AlN (PTA) | 26        |
| In terahertz applications                                                    | 1200           | 1500            | GaAs/InP/Zno           | This work |

Table 2. The comparison with recent works.



**Fig. 10.** Quantum circuit structure by applying two-qubit z stabilizer.



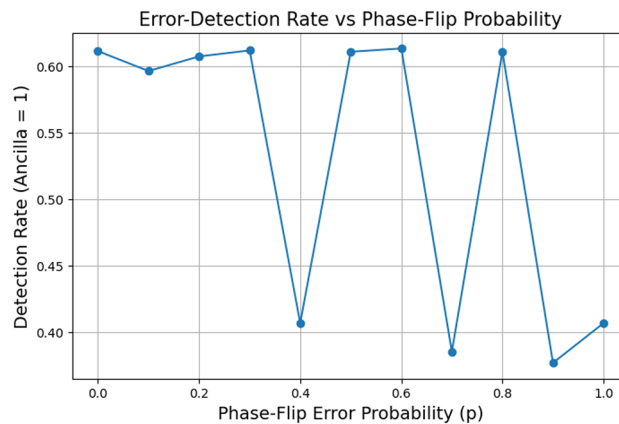
**Fig. 11.** Qubit rotation diagram with phase change of the cNOT gate b. of rotation of qubits with phase changes when phase is applied.

of zinc oxide shows a 40% lower dispersion compared to the indium phosphide material. Since the reduction in dispersion is inversely proportional to the quality factor, therefore, the reduction in dispersion increases the quality factor. This feature is effective in reducing the threshold of the photonic crystal laser. The gain and transmission of the active hybrid materials in the photonic crystal laser are examined below. The mode volume is obtained at a wavelength of 1160 nm and the results show that at this wavelength, considering the refractive index of the cavity material, which is 3.4 for indium phosphide, it is obtained with a ratio of  $1160/(3 \cdot 4)$ . With a larger mode volume, the field distribution becomes wider and the overlap with the output power gain region increases.

The variation of the quality factor with the wavelength of the nanolaser has been investigated. As can be seen, the variation of the quality factor with the wavelength from 1.72 to 1.88  $\mu\text{m}$  increases almost linearly. These variations are obtained between 450 and 1500 for the wavelength of 1.88  $\mu\text{m}$ , so the laser resonance at 1.88  $\mu\text{m}$  is about 3 times more stable than at 1.72  $\mu\text{m}$ . The energy loss is greatly reduced and the photon remains in the cavity for a longer time. The results are compared with the results from recent research in Table 2.

Despite the remarkable progress in photonic crystal nanolasers in terms of mode volume reduction, Purcell factor increase, and laser threshold reduction, these structures have been mainly considered as passive or semi-





**Fig. 12.** Diagram of error detection due to noise changes and phase rota.

active light sources. On the other hand, photonic quantum logic gates have been developed independently, mainly on waveguide-based or interferometric platforms. So far, the direct link between photonic crystal nanolasers as optical active elements and quantum gate-based logic drives at the device level has not been comprehensively investigated.

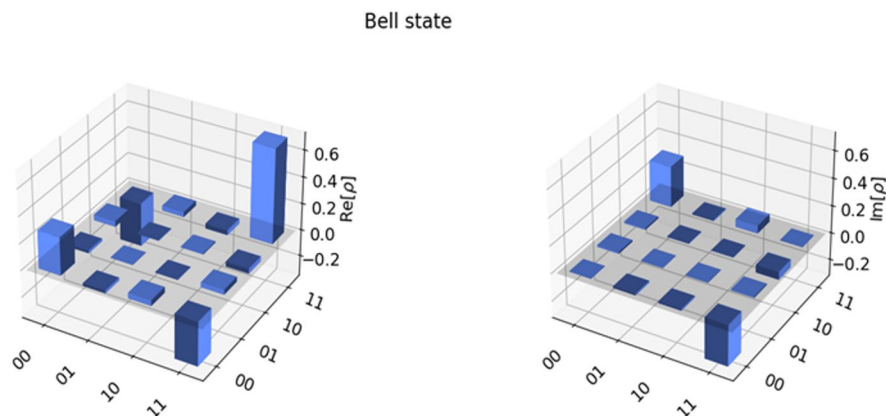
Unlike classical bits, which can only change from 0 to 1, a quantum bit (qubit) can experience more complex errors, including bit shifts, phase shifts, and combinations thereof. The basic principle of error correction is to encode a single logical qubit into a larger number of physical qubits. This redundancy allows errors to be detected and corrected without directly measuring the quantum state, which would destroy it. The overall quantum error correction cycle can be divided into three main steps: Encryption: The logical qubit is encrypted using multiple physical qubits in a protected state. Syndrome measurement: Ancilla qubits are measured to extract information about errors (“syndrome”) without revealing the state of the logical qubit. Decoding and correction: A classical computer uses the syndrome data to identify the most likely error and applies a correction operation to the physical qubits. This process is repeated continuously to maintain the integrity of the logical qubit. The goal is to ensure that the logical qubit error rate is significantly lower than that of the physical qubits. This is known as reaching the “fault tolerance threshold”. Some important challenges in quantum circuits Correlated noise: Most current QEC models assume that qubits error independently. However, in physical systems, an error in one qubit can affect neighboring qubits and create correlated noise. This phenomenon, often caused by shared control lines or thermal effects, can seriously degrade the performance of existing decoders, and new decoding codes and algorithms are needed to handle such errors. Qubit uniformity: Generating a large number of qubits with nearly identical properties (e.g., resonant frequencies, coherence times) is a significant challenge. Variations can lead to different error rates across the chip, making it difficult to implement uniform QEC effectively. Error detection approaches based on phase variations include eight configurations of applying a z error to the data qubit. s error with phase  $\pi/2$  and a phase error with the inverse effect of the s gate. T error with phase  $\pi/4$ . Random error with noise and phase error after the gate. Phase error on the qubit and simultaneous Z error on two qubits. Among these configurations, due to the more predictable error than independent errors and the estimation of joint coherence, the existence of a simultaneous phase error Z on two qubits (Correlated Phase Error) is considered more important than other configurations. Since it is more stable than a single qubit error, multi-qubit cavity superconducting ion trips are important in many photonic Rydberg applications. We have used standard quantum VQE circuits on IBM quantum hardware to characterize and benchmark quantum computers in order to assess the precision of quantum computing noise models. A technique for characterizing an unknown state  $\rho$  is quantum state tomography (QST). To get a sense of how accurate the hardware is, this condition may then be compared to the ideal, anticipated state of a quantum calculation. The process matrix of quantum gates is measured by quantum process tomography (QPT). The preparation and measurement of the state are taken into account in both QPT and QST. By combining a logical CNOT gate and an Anila logic state, Zhang and colleagues were able to implement arbitrary single-state rotation gates. The Pauli logical transition matrices and the average fidelity of all logical gates are used to completely characterize them in a full state set. According to, the input states are found to be 79.0(3)%. Unlike classical bits, which can only change from 0 to 1, a quantum bit (qubit) can experience more complex errors, including bit shifts, phase shifts, and combinations thereof. The basic principle of error correction is to encode a single logical qubit into a larger number of physical qubits. This redundancy allows errors to be detected and corrected without directly measuring the quantum state, which would destroy it. The overall quantum error correction cycle can be divided into three main steps: Encryption: The logical qubit is encrypted using multiple physical qubits in a protected state. Syndrome measurement: Ancilla qubits are measured to extract information about errors (“syndrome”) without revealing the state of the logical qubit. Decoding and correction: A classical computer uses the syndrome data to identify the most likely error and applies a correction operation to the physical qubits. This process is repeated continuously to maintain the integrity of the logical qubit. The goal is to ensure that the logical qubit error rate is significantly lower than that

of the physical qubits. This is known as reaching the “fault tolerance threshold”<sup>25</sup>. Some important challenges in quantum circuits Correlated noise: Most current QEC models assume that qubits error independently. However, in physical systems, an error in one qubit can affect neighboring qubits and create correlated noise. This phenomenon, often caused by shared control lines or thermal effects, can seriously degrade the performance of existing decoders, and new decoding codes and algorithms are needed to handle such errors. Qubit uniformity: Generating a large number of qubits with nearly identical properties (e.g., resonant frequencies, coherence times) is a significant challenge. Variations can lead to different error rates across the chip, making it difficult to implement uniform QEC effectively. Error detection approaches based on phase variations include eight configurations of applying a z error to the data qubit. s error with phase  $\pi/2$  and a phase error with the inverse effect of the s gate. T error with phase  $\pi/4$ . Random error with noise and phase error after the gate. Phase error on the qubit and simultaneous Z error on two qubits. Among these configurations, due to the more predictable error than independent errors and the estimation of joint coherence, the existence of a simultaneous phase error Z on two qubits (Correlated Phase Error) is considered more important than other configurations. Since it is more stable than a single qubit error, multi-qubit cavity superconducting ion trips are important in many photonic Rydberg applications<sup>26</sup>. We have used standard quantum VQE circuits on IBM quantum hardware to characterize and benchmark quantum computers in order to assess the precision of quantum computing noise models. A technique for characterizing an unknown state  $\rho$  is quantum state tomography (QST). To get a sense of how accurate the hardware is, this condition may then be compared to the ideal, anticipated state of a quantum calculation. The process matrix of quantum gates is measured by quantum process tomography (QPT). The preparation and measurement of the state are taken into account in both QPT and QST.

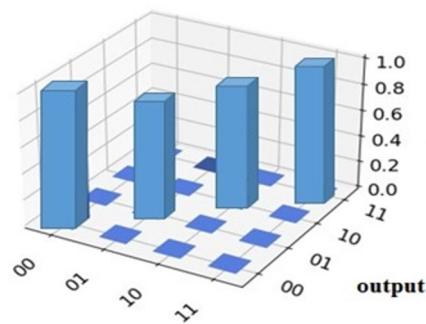
The structure of Fig. 10 is shown by implementing a two-qubit stabilizer. As can be seen, an ancilla qubit is used to detect the phase error after CNOT so that the output is 1 when a phase error Z occurs on  $q_0$  or  $q_1$  or both, and the output is 0 if there is no Z error. and the output is 0 if there is no Z error.

The teleport diagram of the Rydberg scheme has been investigated, as can be seen in the diagram in Fig. 11 shows the Rydberg teleport curve in the noise-free state. The sine wave with a  $90^\circ$  shift between +1 and -1 shows that as  $\theta$  increases, the Bloch vector rotates in the XY plane. Exactly, the sine wave rotates along with X according to a complete circular motion. The x, y paths form a circle with radius 1. The Z curve shows zero, meaning the state never goes up or down because  $R_z$  is just a rotation in the XY plane. In case 11b, in the case of X and Y noise, the reduced amplitude of Z is still close to zero.

As can be seen in Fig. 12, the higher the probability of an error, the higher the rate at which the ancilla syndrome detects it.



**Fig. 13.** Bell-state tomography diagram a) Real and imaginary parts of the reconstructed post-selected density matrix obtained in an entanglement gate operation. The Bell-state fidelity after selection is 69(2)%.



**Fig. 14.** Bell-state tomography for the output, showing an average fidelity of (4)%.

| Logic gate                  | fidelity | Ref       |
|-----------------------------|----------|-----------|
| Rz(0)                       | 96.9(3%) | 29        |
| Rx(0)                       | 75.9%    | 30        |
| Rx( $\pi/2$ )               | 99%      | 31        |
| parallel CZ gate            | 63.09    | 32        |
| Rz( $\pi/4$ ) $\pi/4$ ) Rx( | 99.6(4%) | This work |

**Table 3.** The results are compared with some recent studies.

Pseudo Code in qiskit

```
-----
from qiskit import QuantumCircuit, transpile
from qiskit_aer import AerSimulator
from qiskit.visualization import plot_histogram, plot_state_city
import qiskit.quantum_info as qi
qc=QuantumCircuit(2,2)
#qc.h([0,1])
#qc.cx(0,1)
qc.ry(math.pi/8,0)
qc.rz(math.pi/4,0)
qc.ry(math.pi/8,1)
qc.rz(math.pi/4,1)
qc.cx(0,1)
#qc.ry(math.pi/8,0)
#qc.rz(math.pi/4,0)
#qc.ry(math.pi/8,1)
qc.rz(math.pi/4,1)
qc.measure([0,1],[0,1])
qc.measure(0,0)
qc.measure(1,1)
#qc.measure(2,2)
#qc.draw('mpl')
print(qc)
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```

The quantum process tomography now evaluates the performance using quantum process tomography, rather than merely defining the output state for a small number of chosen input states. It provides a second thorough explanation of the quantum process. In other words, it converts the  $4 \times 4$  density matrix  $\rho_{in}$  at the input to the  $4 \times 4$  density matrix  $\rho_{out}$  at the output. We assume that map  $E$  is linear. As a result, as demonstrated in the Fig. 13, it is enough to experimentally identify  $\rho_{out}$  using quantum state tomography based on the 16 matrices  $\rho_{in}$  in order to define  $E$ . 14. The average fidelity is 99.6%. It is discovered at the point where all error bars represent the statistic's uncertainty of one standard deviation. By employing highly excited Rydberg states, the Rydberg blocking mechanism represents a characteristic of neutral atom systems that allows for strong and programmable two-body interactions. neutral-atom arrays provide a promising architecture for this purpose, thanks to their flexible connectivity and strong, controllable interactions via Rydberg excitations.

One of the features of the Rydberg scheme in the experimental results is that the direct implementation of multi-qubit gates can solve these problems by enabling faster execution with less error-prone operations, ultimately increasing accuracy and reducing overhead.

Therefore, the reconstructed map E can be written in a matrix representation as<sup>27</sup>

$$\epsilon(p) = \sum_{i,j}^{16} A_i p A^t_{x_{i,j}} \quad (3)$$

where the basis A of the operators is created by arbitrarily choosing  $4 \times 4$  matrices  $A_i$ . The  $16 \times 16$  process is made up of the complex coefficients  $x_{(i,j)}$ . matrix x. Naturally, the value of  $x_{(i,j)}$  depends on the base A that is selected. A linear and By inverting a linear system, an unbiased estimate for x is obtained. Since the efficiency function is real Hermitian for all inputs, the complete description of the The efficiency function can therefore be calculated from the 16 input density matrices. The Hermitian function can then be reconstructed by inverting a linear system<sup>28</sup>:

$$\bar{\eta} = \frac{1}{4} \sum_{i=1}^4 \eta(|v_i\rangle\langle v_i|) \quad (4)$$

Bell-state tomography for the output (see Fig. 14), showing an average fidelity of (4)%. All error bars in this paper indicate a statistical uncertainty of 1 standard deviation.

In Table 3, the results are compared with some recent studies.

## Conclusion

In this paper, the hybrid gain medium of quantum dot photonic crystal nanolaser in the increase of the pumping of the var with Indium phosphide, zinc oxide and gallium arsenide materials has been investigated, and the degree of confinement of the var with a quality factor of 1600 for the Indium phosphide layer has been obtained. Next, by applying Gaussian pulses to the logical qubits we investigate the effect of changing the phase-shifting gate R on improving the roundness of the CNOT gate. The results show that in the dipole-dipole interaction between the Rydberg atoms on the optical photons, the excitation between the two resonant states of the input and output photons, the interaction between the two Rydberg states detunes the system from two-photon resonance, resulting in low transmission of the target photon. In general, this leads to the loss of the target photon conditional on the presence of a control photon. To convert the conditional loss into a conditional  $\pi$  phase shift,  $\pi/4$  and  $\pi/8$  conditional phase shift gates are used. The results show that using a phase shifter at the input and output of the CNOT gate conditionally can improve the efficiency by up to 99.6% in the quantum gate structure. The CNOT gate is the only gate that can provide a direct two-qubit interaction for the stabilizer.

## Data availability

All data is available as a reasonable request from corresponding authors.

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## Author contributions

Ali Farmani works on research background, draft and final structure. Anis Omodniaee works on simulation and research.

## Declarations

## Competing interests

The authors declare no competing interests.

## Additional information

**Correspondence** and requests for materials should be addressed to A.F.

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